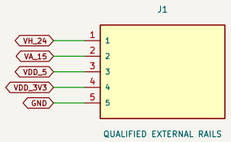
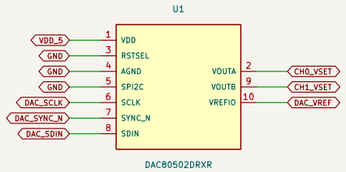
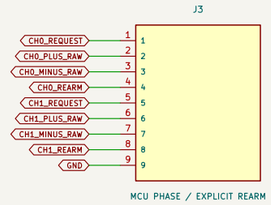
ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.







Source0_analog

File: source0.kicad_sch

Source0_permission

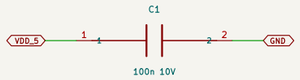
File: permission0.kicad_sch

Source1_analog

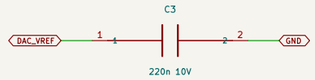
File: source1.kicad_sch

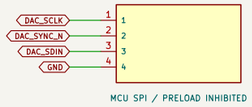
Source1_permission

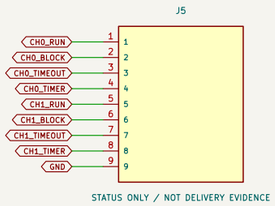
File: permission1.kicad_sch

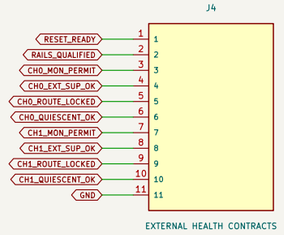












DAC initialization before READY: GAIN[0x04]=0x0103, reference divider 2 and both output buffer gains 2 => nominal 2.5 V FSR.

Code 7864 -> 2.999878 mA nominal through 100 ohm. Zero code is not terminal disconnection. Read README and external contracts.

RESET_READY held LOW through startup; REARM is an explicit pulse with REQUEST/TIMER LOW and independently observed quiescence.











Native circuit slice; artificial loads only; no PCB or implant qualification

Project SCCI RH

Sheet: /

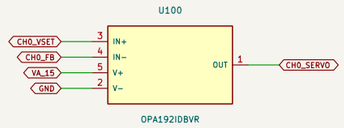
File: current_driver.kicad_sch

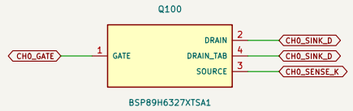
Title: Project SCCI RH S4 / Dual setpoint and external contracts

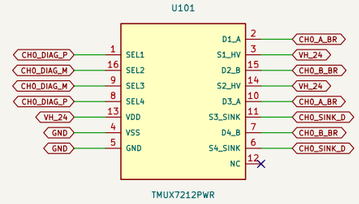
Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/5

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

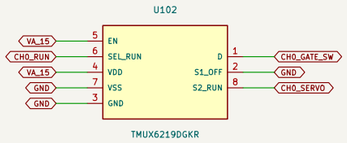
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

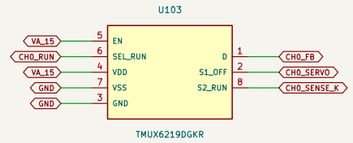


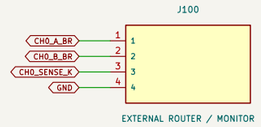




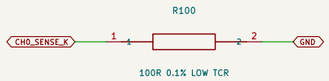
Current path: VH -> upper switch -> artificial pair load + external 2 ohm monitor -> lower switch -> BSP89 -> 100 ohm -> GND.

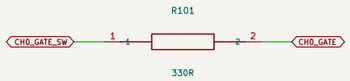


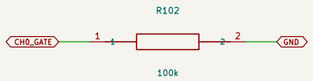


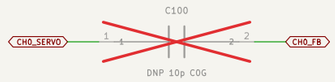


RUN low grounds gate and closes local unity feedback. RUN high selects current loop. DNP compensation needs a transient study.

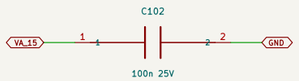


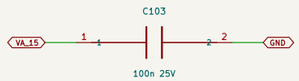






















Bridge uses captured direction; all four commanded off during inhibit. Analog rail clamps, input bias and stuck switches remain real paths.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project SCCI RH

Sheet: /Source0_analog/
File: source0.kicad_sch

Title: Project SCCI RH S4 / Source 0: regulated sink and polarity bridge

Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/5

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

U300
CH0_REQUEST
CH0_T_SET
CH0_T_DIV
VDD_3V3
GND
TRIG_RAW_REQ
SET
DIV
V+
GND
OUT
CH0_TIMER
LTC6993HS6-1#TRPBF

U301
CH0_PLUS_RAW
CH0_MINUS_RAW
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_VALID_DIR
SN74LVC1G86DBVR

U302
CH0_MON_PERMIT
CH0_EXT_SUP_OK
CH0_ROUTE_LOCKED
VDD_3V3
GND
A
B
C
VCC
GND
Y
CH0_HEALTH_A
SN74LVC1G11DBVR

U303
RESET_READY
RAILS_QUALIFIED
CH0_VALID_DIR
VDD_3V3
GND
A
B
C
VCC
GND
Y
CH0_HEALTH_B
SN74LVC1G11DBVR

U304
CH0_HEALTH_A
CH0_HEALTH_B
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_BASE_GOOD
SN74LVC1G08DBVR

U305
CH0_REQUEST
VDD_3V3
GND
A
VCC
GND
NC
Y
CH0_REQ_N
SN74LVC1G04DBVR

U306
CH0_TIMER
VDD_3V3
GND
A
VCC
GND
NC
Y
CH0_TIMER_N
SN74LVC1G04DBVR

U307
CH0_TIMER
CH0_REQ_N
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_STOP_CONSUME
SN74LVC1G08DBVR

U308
CH0_STOP_CONSUME
VDD_3V3
GND
A
VCC
GND
NC
Y
CH0_STOP_N
SN74LVC1G04DBVR

U309
CH0_BASE_GOOD
CH0_STOP_N
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_BLOCK_PRE_N
SN74LVC1G08DBVR

U310
CH0_REARM
CH0_BASE_GOOD
CH0_QUIESCENT_OK
VDD_3V3
GND
A
B
C
VCC
GND
Y
CH0_REARM_A
SN74LVC1G11DBVR

U311
CH0_REARM_A
CH0_REQ_N
CH0_TIMER_N
VDD_3V3
GND
A
B
C
VCC
GND
Y
CH0_SAFE_REARM
SN74LVC1G11DBVR

U312
CH0_SAFE_REARM
VDD_3V3
GND
A
VCC
GND
NC
Y
CH0_LATCH_CLR_N
SN74LVC1G04DBVR

U313
GND
GND
CH0_LATCH_CLR_N
CH0_BLOCK_PRE_N
VDD_3V3
GND
CLK
D
CLR_N
PRE_N
VCC
GND
Q
Q_N
CH0_BLOCK
CH0_BLOCK_N
SN74LVC1G74DCUR

U314
CH0_TIMEOUT
CH0_REQUEST
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_TIMEOUT_D
SN74LVC1G32DBVR

U315
CH0_TIMER_N
CH0_TIMEOUT_D
CH0_LATCH_CLR_N
VDD_3V3
VDD_3V3
GND
CLK
D
CLR_N
PRE_N
VCC
GND
Q
Q_N
CH0_TIMEOUT
CH0_TIMEOUT_N
SN74LVC1G74DCUR

U316
CH0_SAFE_REARM
CH0_PLUS_RAW
RESET_READY
VDD_3V3
VDD_3V3
GND
CLK
D
CLR_N
PRE_N
VCC
GND
Q
Q_N
CH0_PLUS_LATCHED
CH0_MINUS_LATCHED
SN74LVC1G74DCUR

U317
CH0_BASE_GOOD
CH0_BLOCK_N
CH0_TIMEOUT_N
VDD_3V3
GND
A
B
C
VCC
GND
Y
CH0_GOOD
SN74LVC1G11DBVR

U318
CH0_REQUEST
CH0_TIMER
CH0_GOOD
VDD_3V3
GND
A
B
C
VCC
GND
Y
CH0_RUN
SN74LVC1G11DBVR

U319
CH0_RUN
CH0_PLUS_LATCHED
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_DIAG_P
SN74LVC1G08DBVR

U320
CH0_RUN
CH0_MINUS_LATCHED
VDD_3V3
GND
A
B
VCC
GND
Y
CH0_DIAG_M
SN74LVC1G08DBVR

C300 VDD_3V3 GND 100n 10V
C301 VDD_3V3 GND 100n 10V
C302 VDD_3V3 GND 100n 10V
C303 VDD_3V3 GND 100n 10V
C304 VDD_3V3 GND 100n 10V
C305 VDD_3V3 GND 100n 10V
C306 VDD_3V3 GND 100n 10V
C307 VDD_3V3 GND 100n 10V
C308 VDD_3V3 GND 100n 10V
C309 VDD_3V3 GND 100n 10V
C310 VDD_3V3 GND 100n 10V
C311 VDD_3V3 GND 100n 10V
C312 VDD_3V3 GND 100n 10V
C313 VDD_3V3 GND 100n 10V
C314 VDD_3V3 GND 100n 10V
C315 VDD_3V3 GND 100n 10V
C316 VDD_3V3 GND 100n 10V
C317 VDD_3V3 GND 100n 10V
C318 VDD_3V3 GND 100n 10V
C319 VDD_3V3 GND 100n 10V
C320 VDD_3V3 GND 100n 10V

R300 CH0_T_SET GND 175.8k 0.1%
R301 VDD_3V3 CH0_T_DIV 976k 0.1%
R302 CH0_T_DIV GND 182k 0.1%
R303 CH0_REQUEST GND 10k
R304 CH0_PLUS_RAW GND 10k
R305 CH0_MINUS_RAW GND 10k
R306 CH0_REARM GND 10k
R307 CH0_MON_PERMIT GND 10k
R308 CH0_EXT_SUP_OK GND 10k
R309 CH0_ROUTE_LOCKED GND 10k
R310 CH0_QUIESCENT_OK GND 10k

BASE_GOOD=MON & EXT & ROUTE & READY & RAILS & XOR(raw directions). Raw REQUEST clocks TIMER: SAFE_REARM preloads direction while off.

Normal request end consumes BLOCK: TIMEOUT records held request at timer fall. REARM requires request/timer LOW + QUIESCENT_OK.

Static/event checks do not qualify asynchronous hazards, supply ramps, propagation delay or real current tails. No automatic retry.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project SCCI RH

Sheet: /Source0_permission/

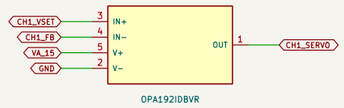
File: permission0.kicad_sch

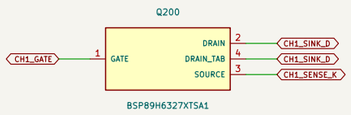
Title: Project SCCI RH S4 / Source 0: independent duration and sticky admis

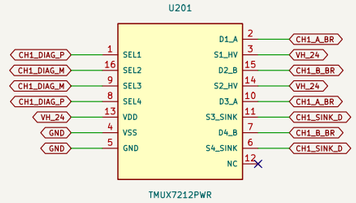
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/5

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

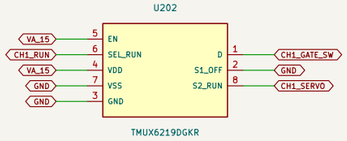
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

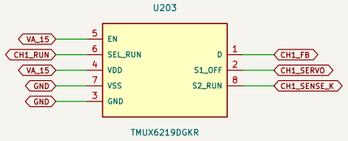


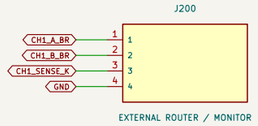




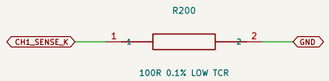
Current path: VH -> upper switch -> artificial pair load + external 2 ohm monitor -> lower switch -> BSP89 -> 100 ohm -> GND.



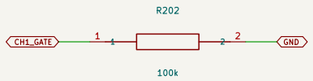


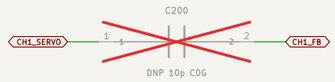


RUN low grounds gate and closes local unity feedback. RUN high selects current loop. DNP compensation needs a transient study.











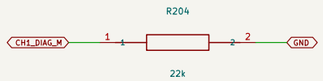














Bridge uses captured direction; all four commanded off during inhibit. Analog rail clamps, input bias and stuck switches remain real paths.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project SCCI RH

Sheet: /Source1_analog/
File: source1.kicad_sch

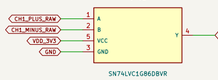
Title: Project SCCI RH S4 / Source 1: regulated sink and polarity bridge

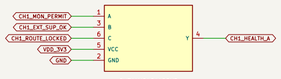
Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/5

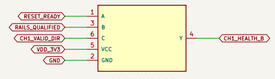
ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.



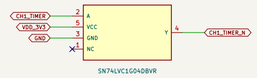








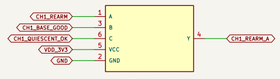






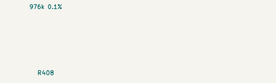


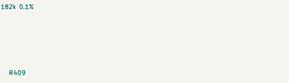




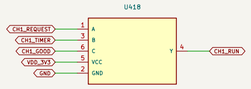


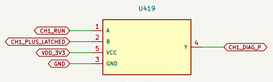


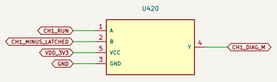












































































BASE_GOOD=MON & EXT & ROUTE & READY & RAILS & XOR(raw directions). Raw REQUEST clocks TIMER: SAFE_REARM preloads direction while off.

Normal request end consumes BLOCK: TIMEOUT records held request at timer fall. REARM requires request/timer LOW + QUIESCENT_OK.

Static/event checks do not qualify asynchronous hazards, supply ramps, propagation delay or real current tails. No automatic retry.

Native circuit slice: artificial loads only; no PCB or implant qualification

Project SCCI RH

Sheet: /Source1_permission/

File: permission1.kicad_sch

Title: Project SCCI RH S4 / Source 1: independent duration and sticky admis

Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/5